

Control work 6 · Quadratic equations

The Chapter III assessment, and the four errors it produces every year.

LESSON 77–78

2 × 40 MIN

ALGEBRA 8, CHAPTER III

STAGE 9 · 4.1

LESSON CLOCK

2'

36'

2'

Setting up	2 min
The paper	36 min
Collect in	2 min

LEARNING OBJECTIVES

- Assess incomplete equations, the discriminant, the formula, Vieta and word problems.
- Diagnose each lost mark by error type.
- Re-solve every task that was lost.

TEXTBOOK

National	Revision of Chapter III
Cambridge	Learner's Book pp. 84–95
Workbook	Workbook 4.1

01 Explanation

Lesson 77 — the paper (40 minutes)

Two variants of seven tasks, 2 marks each:

- task 1 · an incomplete quadratic
- task 2 · solve by factorising
- task 3 · find D and say how many roots
- task 4 · solve with the formula
- task 5 · Vieta — find a missing coefficient or root
- task 6 · a biquadratic equation
- task 7 · a word problem

Marking note: task 7 loses a mark if the impossible root is not rejected in writing.

Lesson 78 — work on mistakes (40 minutes)

THE FOUR ERRORS THIS PAPER PRODUCES

- Sign slips in D** — writing $-4ac$ as negative when c is already negative.
- Dividing by x** , losing the root $x = 0$.
- Negative t kept** in a biquadratic, producing imaginary “roots”.

4. The impossible root not rejected in a word problem.

02 Worked examples

EXAMPLE 1 Find the error: for $x^2 - 2x - 3 = 0$, $D = 4 - 12 = -8$

① $c = -3$, so $-4ac = -4 \cdot 1 \cdot (-3) = +12$.

Two minus signs make a plus.

② $D = 4 + 12 = 16$

③ $x = \frac{2 \pm 4}{2} = 3 \text{ or } -1$

ANSWER

$D = 16$, roots 3 and -1

EXAMPLE 2 Find the error: $x^2 = 4x$, so $x = 4$

① Both sides were divided by x — which may be zero.

② $x^2 - 4x = 0 \implies x(x - 4) = 0$

Factorise instead.

③ $x = 0$ or $x = 4$

The root 0 was lost.

ANSWER

$x = 0$, $x = 4$

03 Interactive model

Show the wrong working, take a vote on the error type, then reveal.

Diagnose the error

Claimed: for $x^2 - 2x - 3 = 0$, $D = 4 - 12 = -8$

① $a = 1, b = -2, c = -3$

Read
the
signs
off
carefully.

② $-4ac = -4 \cdot 1 \cdot (-3) = +12$

The
two
minus
signs
cancel.

③ $D = 4 + 12 = 16 > 0$

Two
roots,
not
none.

④ $x = 3$

or $x = -1$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Discriminant	Diskriminant	Дискриминант
Root	Ildiz	Корень
Reject	Rad etish	Отбросить
Check by substitution	O'rniga qo'yib tekshirish	Проверка подстановкой
Word problem	Matnli masala	Текстовая задача

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

For $x^2 - 2x - 3 = 0$, D is:

-8

16

4

-4

$x^2 = 4x$ has roots:

$x = 4$

$x = 0, 4$

$x = \pm 2$

$x = 0$

In a biquadratic, $t = -4$ gives:

$x = \pm 2$

no real x

$x = 2$

$x = -2$

A width comes out as 5 or -8 . The answer is:

both

-8

5, rejecting -8

neither

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. **Variant 1, task 1.** Solve $x^2 - 36 = 0$

ANSWER $x = \pm 6$

2. **Task 2.** Solve $x^2 - 6x + 8 = 0$

ANSWER $x = 2, x = 4$

3. **Task 3.** Find D for $x^2 + 2x + 5 = 0$ and say how many roots.

ANSWER $D = -16$, none

4. **Task 4.** Solve $2x^2 - 5x + 2 = 0$

ANSWER $x = 2, x = 0.5$

5. **Task 5.** One root of $x^2 + bx - 12 = 0$ is 4. Find b .

ANSWER other root -3 , $b = -1$

6. **Task 6.** Solve $x^4 - 5x^2 + 4 = 0$

ANSWER $x = \pm 1, \pm 2$

7. **Task 7.** A rectangle is 2 m longer than wide, area 35 m². Find its sides.

ANSWER $5\text{ m} \times 7\text{ m}$

Medium · the standard question

7 PROBLEMS

1. **Variant 2, task 1.** Solve $3x^2 - 12x = 0$

ANSWER $x = 0, x = 4$

2. **Task 2.** Solve $x^2 + x - 20 = 0$

ANSWER $x = 4, x = -5$

3. **Task 3.** Find D for $2x^2 - 4x + 2 = 0$ and say how many roots.

ANSWER $D = 0$, one

4. **Task 4.** Solve $3x^2 + 5x - 2 = 0$

ANSWER $x = \frac{1}{3}, x = -2$

5. **Task 5.** The roots of $x^2 + px + q = 0$ are 2 and -7 . Find p and q .

ANSWER $p = 5, q = -14$

6. **Task 6.** Solve $x^4 - 10x^2 + 9 = 0$

ANSWER $x = \pm 1, \pm 3$

7. **Task 7.** Two consecutive whole numbers have product 210. Find them.

ANSWER 14 and 15

Hard · stretch and reason

7 PROBLEMS

1. Find the error: for $x^2 - 2x - 3 = 0$, $D = 4 - 12 = -8$

ANSWER $-4ac = +12$; correct $D = 16$, roots 3, -1 .

2. Find the error: $x^2 = 4x$, so $x = 4$

ANSWER Dividing by x lost the root 0.

3. Find the error: $x^4 + 3x^2 - 4 = 0$ gives $x = \pm 1, \pm 2$

ANSWER $t = -4$ must be rejected; only $x = \pm 1$.

4. Find the error: a width comes out 5 or -8 , answer “5 and -8 ”

ANSWER A length cannot be negative — the rejection must be stated.

5. Find the error: “ $x^2 = 16$, so $x = 4$ ”

ANSWER The negative root was lost: $x = \pm 4$.

6. Find the error: “ $(x - 1)(x - 4) = 6$, so $x - 1 = 6$ or $x - 4 = 6$ ”

ANSWER The zero product rule needs a zero on the right; expand and re-solve.

7. Find the error: “ $D = 25$, so $x = \frac{-b \pm 25}{2a}$ ”

ANSWER The formula uses $\sqrt{D} = 5$, not D itself.

07 Homework

After the work-on-mistakes lesson

Re-solve every task you lost marks on. Quarter IV opens with statistics.

- Solve $2x^2 - 7x - 4 = 0$
- Solve $x^4 - 13x^2 + 36 = 0$
- One root of $x^2 + bx + 15 = 0$ is 5. Find b and the other root.
- A rectangle is 4 cm longer than wide and has area 96 cm^2 . Find its sides.

5. Write out the four errors from this lesson with one example each.